EMATM0067
Introduction to AI and Text Analytics Coursework

Spring 2026. Taught by: Edwin Simpson, Conor Houghton, Karina Nurlybayeva

Deadline: 13.00 on Tuesday 5th May

Estimated effort: approximately up to 70 hours per student.

Group size: five to six students.

Weighting: group report = 80%, individual reflection = 20%.

Overview

For this coursework, you will each be assigned to a team of five to six students. Your group will work on one of four different AI & text analytics tasks. Each task is designed to:

- Develop your ability to design and implement real-world pipelines for analysing text data using AI methods
- Encourage hypothesis-driven exploration and careful evaluation over model complexity
- Provide an opportunity to work in a team to produce more powerful results.

Working as a team, you will implement an analysis pipeline, experiment with key design choices and analyse different aspects of the data, to produce a *group report* describing your method, results and insights. You will also write a short *individual reflection,* documenting your personal contributions to the project.

The tasks have been carefully selected to represent similar levels of difficulty. Each group will be randomly assigned to one of the tasks below, with assignments shared on Blackboard. We will not be able to change task or team assignments based solely on personal preferences, but please talk to your lecturers if you are concerned about your assignment.

Supervision

Your group will be assigned a supervisor who will meet with you at three different points. Their role is not to provide technical advice on solving your task, but to advise on group working and help to ensure all members of the team play a fair part.

Each meeting will be 20 minutes and your supervisor will ask:

1. How has each person contributed since the last meeting?
2. What challenges are you facing and could you overcome them?

3. What objectives and responsibilities does each team member have for the next stage of work?

Supervision points: supervision will take place in the following weeks during the labs.

- Week 21 (before Easter vacation)
- Week 22 (after Easter vacation)
- Week 24 (one week before submission).

Tasks

Task 1: Medical Question Answering

Question-answering systems can help biomedical experts to quickly find answers to important questions, without trawling through mountains of publications or medical records.

Objective: analyse how different text representations and AI methods affect the quality of answers to medical questions produced by question answering systems.

Use the PubMedQA dataset, consisting of questions, abstracts and labelled answers. You can find more information about this task on the task website. A script for loading the data will be provided in the Text Analytics Github repository.

Student group work:

1. Design and implement **a QA pipeline:**
   a. Identify the key components, e.g., question representation, answer generation or classification.
   b. Choose two important design *axes* to explore, e.g., text representation (e.g., embeddings vs. n-grams) or answer classification versus LLM-based generation.
   c. For each design axis, compare 2-3 alternative approaches, motivated by a clear hypothesis about how each approach will perform.
2. Evaluate each variant of your pipeline in terms of:
   a. Answer quality
   b. Error analysis, giving an understanding of specific failure modes.
3. Is it possible to predict answer quality from characteristics of the question? Consider whether regression or a small neural network can decide if a question is likely to produce a good quality answer.
4. Discuss how each design choice affects performance and why.

Task 2: Structured Information Extraction

By converting unstructured descriptions of clinical trials to structured tables, we can help biomedical researchers to locate and synthesise valuable evidence across many different studies.

Objective: Given a corpus of clinical trial abstracts, extract information into a predefined table schema. Evaluate how different extraction strategies affect the quality of the extracted data.

The text data and table schema are provided by the EBM-NLP dataset. The dataset consists of clinical trial abstracts, and the task is to extract the *population, intervention, comparator* and *outcomes* of the trial from the abstract. A script for loading the data will be provided in the Text Analytics Github repository.

Student group work:

1. As a preliminary consider **clustering** approaches: Before extraction, use k-means or Hierarchical clustering (HAC) on sentence embeddings to see whether natural clusters correspond to schema fields.
2. Design and implement **an information extraction pipeline**:
    a. Choose two important design *axes* to explore, such as: rule-based approaches versus LLMs; end-to-end models versus decomposing the task into smaller steps; varying training and zero/few-shot learning strategies.
    b. For each chosen approach, design a complete pipeline for extracting the tabular data.
    c. For each design axis, compare 2-3 alternative approaches, motivated by a clear hypothesis about how each approach will perform.
3. Evaluate each variant of your pipeline in terms of:
    a. Field-level accuracy/F1/etc.
    b. Coverage vs precision,
    c. Downstream usability (e.g., can queries be answered).
4. Discuss trade-offs between approaches and the limitations of your approach.

Task 3: Comparative Corpus Analysis

Analysing changes in scientific abstracts over time can provide data-driven insights into the changing priorities and methods of researchers. These abstracts communicate how researchers frame problems, and the kind of claims that they highlight, as well as trends in applying specific methods.

Objective: Given a corpus of scientific abstracts, analyse how the focus, framing and communication of research has changed within the fields of AI, machine learning and NLP (you can either treat these as a single field or pick one to focus on). Analyse how different text analytics methods affect the insights that can be drawn.

The dataset is available at the [HuggingFace page] and contains metadata such as title, author and journal. The webpage provides further documentation and [the Text Analytics Github repository] provides a script for loading a suitable corpus for your project.

Student group work:

1. Design and implement **a text analysis pipeline** to answer the above question:
    a. Define the key steps in the pipeline, e.g., splitting corpus into periods, preprocessing, text representation (TF-IDF vectors, embeddings, etc.), labelling or describing each period, comparing periods.
    b. Choose two important analytic *axes* to explore, such as: text representation (e.g., words, phrases, topics, embeddings); comparison method (keyword analysis, distance measures, classifying text into periods); temporal granularity; preprocessing choices.
    c. For each design axis, compare 2-3 alternative approaches, motivated by a clear hypothesis about how each approach will perform.
2. Examine the dates where temporal drift occurs by plotting abstracts in a **dimensionally reduced space**. Apply PCA to TF-IDF or embedding space. Project abstracts over time and visualise the trajectory of the field in the low-dimensional space.
3. Present your main findings from the corpus in answer to the comparison question.
4. Evaluate each tested approach in your pipeline in terms of:
    a. Consistency and robustness of findings across methods
    b. Interpretability of output

    c. Alignment with known or plausible patterns.
5. Discuss trade-offs between approaches and the limitations of your approach.

Task 4: Mining Insights from Customer Feedback

Text analytics can help organisations to uncover hidden problems, emerging concerns, or strengths of their products and services that customers express in support tickets.

Objective: Analyse a corpus of customer support tickets to identify and summarise the main issues raised by customers. These issues should provide actionable insights that could inform improvements to products or services. Evaluate how different text analytics methods affect the insights you extract.

You can find more information about this task on the website and a script for loading the data in the Text Analytics Github repository. If you prefer, you could use a larger, alternative dataset of customer support tickets.

Student group work:

1. Look at **clustering** of issue type by using k-means or HAC on TF-IDF vectors or Embeddings. Examine cluster coherence and interpretability and investigate stability across k.
2. Design and implement **a text analytics pipeline** to identify the main kinds of issue raised by customers**:**
    a. Identify the key steps, e.g., exploring data distributions, preprocessing, text representation (TF-IDF vectors, embeddings, etc.), labelling or discovering specific issues/themes, ranking issues by importance.
    b. Choose two important analytic *axes* to explore, e.g., text representation (e.g., words, phrases, topics, embeddings); labelling methods (e.g., for sentiment analysis); topic models or methods for summarising issues/themes; temporal trends.
    c. For each analytic axis, compare 2-3 alternative approaches, motivated by a clear hypothesis about how each approach will perform.
3. Present your main findings from the corpus to show actionable insights.
4. Evaluate each tested approach in your pipeline in terms of:
    a. Consistency and robustness of findings across methods
    b. Coherence and interpretability of issue categories
    c. Alignment with intuitive or expected patterns and with the initial clustering analysis.
5. Discuss how each design choice affects performance and why.

Task 5: Build a Neural Network to Describe Simple Pictures.

Relating visual inputs to natural language descriptions is a core challenge in AI. This task involves working with multimodal data to explore how simple neural networks can be adapted to generate image descriptions.

Objective: The goal of this task is to see if a simple neural network can describe simple pictures; it is intended that the task use generated images so that there is an abundance of labelled data.

You will find sample code to help you get started on this task in the Text Analytics Github repository.

Student group work:

1. Dataset generation – write a script to generate sentences based on some simple relationships. Here are some examples:

a. Arrangements of simple shapes, such as "a big blue sphere is above a small red cube"; by reducing each part of the sentence to a simple symbol, it should be easy to generate a large number of sentences. For example, define a list of objects {o_1, o_2, . . ., o_n} and colours {c_1, c_2, . . ., c_m}, then use these to complete *nm* two-word colour-object phrases by selecting objects *o_i* and colours *c_j* to complete a sentence template. Ideally, the sentences won't all have the same structure.
b. Make images of numbers of varying length and sizes with different colours – "large blue 67", "small red 1337" and so on.
c. Make tic-tac-toe boards in different game states corresponding to different sentences describing the game, e.g., "X has gone in the center, 0 in the top left and bottom right, X in the middle top".

2. Write a script to generate images from your symbol sequences. For 1a and 1c, you could use a Python graphics library to draw shapes or draw Xs and 0s on a grid. For 1b use, for example, MNIST to draw different versions of each number.
3. Design and implement **a neural network for mapping images to sentences:**
   a. Text representation: convert the sentences to a suitable representation, e.g., use a language model to obtain sentence embeddings.
   b. Map images to text: build a CNN (or similar) to predict the sentences from the pictures.
   c. Compare your approach to a commercial LLM
   d. Choose a further analytic *axis* to explore, such as: what text representation or CNN architecture variant is most effective?
   e. For your chosen analytic axis, compare 2-3 alternative approaches, motivated by a clear hypothesis about how each approach will perform.
4. Evaluate each method in terms of:
   a. Sentence prediction accuracy – does your method work?
   b. Does a commercial LLM do better than your neural network?
   c. Error analysis, giving an understanding of performance variation with different sentence or image structures
5. Discuss how each design choice affects performance and why.

Group Report (80%)

Your team's report should be structured into the headings described below. The descriptions below are guidelines, rather than strict rules, and each section should include all the information that you think is relevant.

1) Abstract (0.25 pages): a brief summary of your project, stating the main challenge, the ideas you tested, and key findings from the experiments.
2) Introduction (1.5 pages): outline the task you have been assigned, its motivations, the available data and requirements for a good solution. You may wish to include examples from the dataset and/or some data exploration plots.
3) Methods (3.5 pages):
   a) Explain the baseline system, including important preprocessing steps and chosen text representations or features. Include figures and/or equations to make your design clearer.
   b) Discuss the axes of experimentation, how they could affect performance.
   c) Describe your proposed improvements along each chosen axis. Include hypotheses about why you think these improvements will help.
4) Experimental evaluation (3.5 pages):

a) Explain your choice of performance metrics and their limitations.
b) Describe your experimental setup: how each dataset split was used, how hyperparameters were set, and any other important information about how your methods were run. It is not necessary to list all the software packages you used, but reference any libraries used for the core parts of your system (e.g., links to HuggingFace models).
c) Present your results in suitable plots and/or tables, providing interpretation of the results to highlight important observations about the comparative performance of your approaches.
d) Analyse the errors that your methods make to understand if there are particular text patterns that cause more frequent errors.
e) Interpret and discuss your results, explaining the strengths and limitations of each method, and insights or recommendations based on your analysis.

5) Conclusions (1 page): describe the key findings from your experiments, explaining what worked well, what did not work, and the open challenges for this task.

What we are looking for

Motivated design choices, supported by evidence. We are *not* looking for novel hypotheses, state-of-the-art leaderboard scores, or ground-breaking new research. A good report will justify design decisions, with data exploration and experimental analysis to provide support.

A few insightful claims about the nature of the task, and what works well, presented as a coherent story.

Depth, rather than breadth – a report that provides a few interesting insights, with sound experimental results, rather than testing a large number of different models and configurations in a superficial way.

Evidence of understanding the application itself – the requirements for a good system and how your system compares.

Understanding of this particular task or dataset – how its peculiarities may affect the design of your pipeline and the performance of AI and NLP methods.

10% of marks will be given for the overall clarity of the report and how well it communicates your ideas, analysis and results.

Group Report Formatting

- Maximum 10 pages of A4 for the group task
  - Font must be at least 11pt, margins at least 2.5cm/1 inch, with single line spacing.
  - Do not try to squeeze extra content in – you will lose marks. Conversely, there will be higher marks for more concise writing.
  - References do not count toward the page limit.
- Use the Overleaf template from either NeurIPS or ACL to format your report (or the MSWord equivalent). This will automatically set margins and fonts.
- The text in your figures must be big enough to read without zooming in.
- Use citations where appropriate, following a typical citation and reference style, as in the NeurIPS or ACL guidelines above.

Individual Reflection (20%)

The individual reflection is intended to help us understand your contributions to the group work, as well as your own understanding of the assigned tasks and your proposed solution.

Your individual reflection should use the same formatting as the group report, and include the following sections (available marks in brackets):

1. A brief summary of your contributions, a visualisation of your individual activity on Github and your project management tool (max 1 page A4, max 150 words; 8 marks)
2. Technical understanding (max 300 words; 7 marks):
   a. Explain one technical challenge you worked on personally.
   b. How did your contribution affect the results of the group project?
3. Teamwork (max 200 words; 5 marks): How the team coordinated their work and ensured that every member was included. Explain the roles of each team member who contributed and mention any tools that helped you coordinate your teamwork.

Submission

- Each group must submit a report to the submission point named "Group Report – Turnitin Submission Point". Make sure your report includes a link to a repository of your team's code (e.g., a public Github link or OneDrive link that is readable by members of the university).
- Each person must submit their own individual reflection to the submission point "Individual Reflection – Turnitin Submission Point"

Avoiding Academic Offences

Please re-read the university's plagiarism rules to make sure you do not break any rules. Academic offences include submission of work that is not your own, falsification of data/evidence or the use of materials without appropriate referencing. Note that sharing your report with others is also not allowed. These offences are all taken very seriously by the University.

Do not copy text directly from your sources – always rewrite in your own words and provide a citation.

Work independently – do not share your code or reports with others.

Do not use AI to generate parts of your report – this includes automatically translating passages of text from another language.

Suspected offences will be dealt with in accordance with the University's policies and procedures. If an academic offence is suspected in your work, you will be asked to attend an interview, where you will be given the opportunity to defend your work. The plagiarism panel can apply a range of penalties, depending on the severity of the offence. These include a requirement to resubmit work, capping of grades and the award of no mark for an element of assessment.

Extensions and Exceptional Circumstances

If the completion of your assignment has been significantly disrupted by serious health conditions, personal problems, or other serious issues, you can apply for consideration in accordance with university policies. Refer to the guidance and complete the application forms as soon as possible. Please see the guidance below and discuss with your personal tutor for more advice:

https://www.bristol.ac.uk/students/support/academic-advice/assessment-support/request-a-coursework-extension/

https://www.bristol.ac.uk/students/support/academic-advice/assessment-support/exceptional-circumstances/